

ISOM 678 Final Project

The time has come for you to start working on your final project. There are five data sets you can choose from. Please choose **one of them** to work on. Check with me first if you'd like to work on a different problem.

You may work in a group of **up to 3**. Each group is required to deliver a 5-minute final presentation on **Friday, April 24**. During the presentation, you will show your analysis, findings, and recommendations. Assume you are presenting to management that has some understanding of the topics in this class. Please read your task for each data set careful.

You'll also need to submit the following to canvas by **April 27**:

1. Slides - the slides from your presentation. They must be well organized and self-contained, with clearly labelled tables and figures.
2. Code - clean, organized and clearly commented R/Python code, so that I can reproduce your analysis. ALL results shown on your slides must be reproducible from your code.
3. Survey - confidential peer evaluation. Your grade for the final project will be your group grade adjusted by your peer evaluation score. For example, if your group grade is 90 and your peer evaluation score is 90, then your grade for the final project will be 81.

The final project will be graded based on correctness and completeness of the analysis, your understanding of the analysis, and quality of the presentation and slides. This project is similar to tasks you'll soon face in real life. Have fun!

(I) Loan repayment

A lender is interested in understanding how to best predict loan repayment. Your task is twofold with:

- (T1) Inference: to help them understand the risk of non-repayment and propose a targeting plan for future loans;
- (T2) Prediction: to build a predictive model with high prediction accuracy. For this task, you can create your own training and testing data split for prediction accuracy evaluation. Please also evaluate the performance of a deep learning model.

The data include information about loan information such as Loan Type, FICO, Years In Business,

Your response variable of interest is **PRSM** (performance ratio at six months). Note that this column is not included in the file but can be computed as

$$PSRM = 2 \times \frac{\text{Amount repaid at six months}}{\text{Total amount to be repaid}}.$$

PRSM should be approximately equal to 1 if the payments at 6 months are on track to fulfill the debt at the end of the year. Values of $PRSM < 1$ indicate a loan for which the payments are currently coming in slower than expected; $PRSM > 1$ indicates a loan that is being paid off faster than expected.

You can read in the data and compute PRSM as follows.

```
loans <- read.csv('https://zhang-datasets.s3.us-east-2.amazonaws.com/PRSM.csv')
loans$PRSM<-2*as.numeric(loans$Amt.Repaid.at.6.Months)/loans$Total.Amt.to.be.Repaid
```

Discussions with the lender have suggested several subtle issues that should be addressed when modeling these data. These comments from the lender may suggest variables that will be useful to you in modeling the performance of loans.

- (a) Merchants that seek loans from your client often operate in distressed neighborhoods. The lender believes that improvements to the economic condition of the neighborhood (income, jobs, housing, etc) provide an environment in which the merchant will more easily be able to keep up with the target payment stream.
- (b) The lender feels that some independent service organizations (ISOs) provide much better (or much worse) customers than others. The lender would very much like some evidence to either support or contradict this suspicion regarding differences among ISOs. Are there particularly good or bad ISOs among those represented in your data?
- (c) There is a strong belief that overly aggressive commissions could be a red flag because the independent sales reps may have a propensity to push such loans, an example of the principal-agent problem.
- (d) Lenders commonly make use of information from a credit bureau, such as the FICO score, as a means to judge the ability of a borrower to repay a loan. This information

is believed to be particularly useful when dealing with new loans but may not be so useful for repeat loans.

- (e) There has been much discussion within the lender of the relevance of the tradelines variable. Some argue that having many tradelines is a good sign in that it proves that the merchant has been actively seeking credit and is a bona fide entity, while others suggest not needing tradelines is a positive sign because the merchant has not required credit to operate.
- (f) Past performance is often a good indicator of future success (or failure). It is hard to believe that delinquencies against the merchant would not in some way be associated with loan performance.

(II) Customer churns

A telecommunications company is concerned about the number of customers leaving their landline business for cable competitors. Your task is twofold with:

(T1) Inference: to help them understand what kind of customers are leaving and propose a retention plan to decrease churn and improve revenues;

(T2) Prediction: to build a predictive model with high prediction accuracy. For this task, you can create your own training and testing data split for prediction accuracy evaluation. Please also evaluate the performance of a deep learning model.

The data can be read in here

```
churn <- read.csv('https://zhang-datasets.s3.us-east-2.amazonaws.com/telcoChurn.csv')
```

and includes information about:

- If a customer left - the column is called Churn (i.e., Churn=1 means already left)
- How long they've been a customer - the column is called tenure
- Services that each customer has signed up for - phone, multiple lines, internet, online security, online backup, device protection, tech support, and streaming TV and movies
- Customer account information - contract, payment method, paperless billing, monthly charges, and total charges
- Demographic info about customers - gender, age range, and if they have partners and dependents

Your response variable of interest is **Churn**.

Note: when you consider variables to include in the model, be careful of the nested ones. For example, **PhoneService** shows whether an account has phone service and **MultipleLines** shows whether an account has multiple lines if it has phone service. So it does not make sense to include both of them in the model because the information in **PhoneService** is nested in **MultipleLines**. Same thing with **InternetService** and **OnlineSecurity**, **OnlineBackup**, **DeviceProtection**, **TechSupport** and **StreamingMovies**, in that these additional add-on features are only relevant for accounts that have **InternetService**.

(III) Walmart sales forecasting

Walmart has provided data weekly sales from 45 stores during 2010-02-05 to 2012-11-01. They are looking to understand the impact of holidays on store sales and have included four holidays weeks into the dataset, including Christmas, Thanksgiving, Super bowl and Labor day. Your task is twofold with:

(T1) Inference: to help identify factors that significantly impact sales;

(T2) Prediction: to build a predictive model for weekly store sales with high prediction accuracy. For this task, please use the **last 30 weeks** as testing data and use the MSE for prediction error evaluation. Please also evaluate the performance of a deep learning model.

There are three datasets provided by Walmart: The data can be read in here

```
stores <- read.csv('https://zhang-datasets.s3.us-east-2.amazonaws.com/stores.csv')
sales <- read.csv('https://zhang-datasets.s3.us-east-2.amazonaws.com/sales.csv')
features <- read.csv('https://zhang-datasets.s3.us-east-2.amazonaws.com/features.csv')
```

- stores.csv: it has three columns. Store: stores numbered from 1 to 45, Size : store sizes, type: store types — A,B and C.
- sales.csv: it has five columns. Store: the store number, Dept: the department number, Date: the week, Weekly_Sales: sales for the given department in the given store, IsHoliday: whether the week is a special holiday week.
- features.csv: it has eleven columns. Store: the store number, Date: the week, Temperature: average temperature in the region, Fuel_Price: cost of fuel in the region, Markdown1-5: anonymized data related to promotional markdowns that Walmart is running. Markdown data is only available after Nov 2011, and is not available for all stores all the time. CPI — the consumer price index, Unemployment: the unemployment rate, IsHoliday: whether the week is a special holiday week.

(IV) Online retail

A UK-based online retail company that sells unique all-occasion gifts wants to better understand customer behavior and improve its product recommendation system. The company has shared a transaction log containing over 500,000 records from December 2010 to December 2011. See information here: <https://www.kaggle.com/datasets/jihyeseo/online-retail-dataset-from-uci-ml-repo>.

Your task is twofold:

(T1) Exploratory data analysis and extraction of the most relevant business insights in the dataset. This may include summaries of product-level characteristics and customer-level characteristics.

(T2) Unsupervised learning: perform customer segmentation using clustering techniques (e.g., K-means) or Autoencoder. Using these segments, provide actionable recommendations for improving customer retention and targeting.

For customer segmentation, it is common to do a RFM (Recency-Frequency-Monetary) analysis. That is, segment customers based on recency (i.e., time of most recent transaction), purchase frequency, and total spending. You are, of course, free to explore other type of clustering analysis.

(T3 Optional) Customer churn prediction: make prediction for customer churns based on the first nine months of data (label: 1 if customer is inactive in the last three month, 0 otherwise). For this task, you can create your own training and testing data split for prediction accuracy evaluation.

The data can be read in here:

```
retail <- read.csv("https://zhang-datasets.s3.us-east-2.amazonaws.com/Online+Retail.csv")
```

(V) Hospital readmissions A hospital system is concerned about the number of diabetic patients who are readmitted within 30 days of discharge. Readmissions are costly and often indicate gaps in care, discharge planning, or follow-up. See data information here: <https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008>. More information on the IDs can be found here: https://archive.ics.uci.edu/ml/machine-learning-databases/00296/dataset_diabetes/diabetic_data.csv.

Your task is twofold:

(T1) Inference: help the hospital understand which factors are associated with a higher risk of 30-day readmission and propose actionable strategies to reduce readmissions and improve patient outcomes.

(T2) Prediction: build a predictive model to identify patients at high risk of 30-day readmission at the time of discharge. You may create your own training and testing data split for evaluation. Compare multiple models, including both traditional statistical models, machine learning models (optional) and deep learning models.

The data can be read in here:

```
hospital <- read.csv("https://zhang-datasets.s3.us-east-2.amazonaws.com/diabetic_data.csv")
hospital$readmit_30 <- ifelse(hospital$readmitted == "<30", 1, 0)
```